Time Series Remote Sensing Image Classification Using Feature Relationship Learning

Peng Dou, Chunlin Huang, *Member, IEEE*, Weixiao Han, Jinliang Hou, *Member, IEEE*, and Ying Zhang

***Abstract*— Recently, time series remote sensing image (TSRSI) has been reported to be an effective resource for mapping fine land use/land cover (LULC), and deep learning, in particular, has been gaining growing attention in this field. However, existing deep learning methods often only learn features from either the temporal or spatial domain, neglecting the intercorrelation between temporal features, which may provide more information for classification, are not fully considered. To make full use of the relations between temporal features and to explore more objective features for improving classification accuracy, we proposed a feature relationship-based classification method. The method leverages the angles between features on the temporal curve to establish relationships between every pair and triplet of features, resulting in the creation of feature relationship matrices (FRMs) and feature relationship tensors (FRTs). Afterward, a 2-D–3-D multiscale convolutional neural network (2-D–3-D MSCNN) was designed to learn deep features from FRM and FRT, achieving the classification improvement of TSRSI. Our experiment was conducted on TSRSIs located in two counties, Sutter and Kings in California, USA. The experimental results indicate that compared to both deep learning and nondeep learning methods, the proposed approach achieves significant improvements in accuracy and LULC mapping, validating the effectiveness and feasibility of enhancing TSRSI classification accuracy through feature relationship learning.**

***Index Terms*— Deep learning, feature relationship, land use and land cover (LULC), remote sensing image classification, time series image classification.**

I. Introduction

OVER the years, the primary method of producing land use and land cover (LULC) maps has been to use machine learning to assign a unique class label to each pixel in single-scene remote sensing images [1], [2], [3]. Several classification methods, including support vector machines (SVMs), naive Bayes, random forests (RFs), multilayer perceptron, and convolutional neural networks (CNNs), have played a significant role in improving the accuracy of LULC classification [4], [5], [6], [7]. However, single-scene images only capture instantaneous spectral characteristics of land features, limiting the amount of information available for classification. This limitation results in a smaller range of identifiable classes and subpar classification performance for time-varying land features such as crops and different types of vegetation, making it challenging to meet the demands for higher precision LULC mapping [8], [9], [10]. Moreover, single-scene images are affected by factors such as seasonality and weather, making their classification results unable to accurately reflect the stable state of LULC. Consequently, this hinders the ability to analyze and adapt to more complex land surface changes on an annual basis [11], [12].

The evolution of an object over time can be a crucial factor in enhancing its recognizability [13]. Time series remote sensing images (TSRSIs) provide a valuable solution to the challenges mentioned above. These images involve multiple remote sensing captures of the same area using the same sensor at different times, forming a data cube comprising of diverse data types such as multispectral and radar images [12], [13], [14], [15], [16]. The classification of TSRSIs involves utilizing continuous observations from these images and categorizing the land surface based on temporal information [17]. In addition to spectral information, TSRSI includes valuable time series data. Furthermore, they provide essential spatial context, as adjacent images exhibit spatial correlation and continuity, offering additional contextual insights [18], [19]. Compared to single-scene images, these characteristics give TSRSI-based classification a distinctive advantage, enabling more precise land feature classification and effectively reducing classification uncertainties. Research on the classification of TSRSIs has found applications in various fields [20], [21]. Notably, fine-grained crop classification stands out as a significant area of focus and presents a key application for temporal remote sensing image classification [10], [22].

According to the composition of temporal features, conventional methods for temporal remote sensing image classification can be broadly categorized into two types [8], [10], [17]. The first approach involves overlaying images from a certain time period and using methods such as SVM to classify and extract LULC information. This method has proven effective in generating annual LULC products. However, aggregating images in the time domain can lead to the "curse of dimensionality" and a significant amount of redundant information [17], [23], [24]. This becomes particu-

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larly evident when utilizing high temporal resolution images (e.g., Sentinel-1 or 2) to construct yearly temporal remote sensing images, which negatively impacts the accuracy of classification [25], [26]. The second approach entails converting remote sensing images into specific indices [e.g., normalized difference vegetation index (NDVI) and soil adjusted vegetation index (SAVI)] and then constructing temporal images for classification purposes. This method allows for the creation of more pronounced temporal feature curves, which play a crucial role in improving the extraction accuracy of different vegetation types and crops [17], [26], [27]. However, due to factors such as parameter settings, feature selection, and noise, this approach necessitates extensive preprocessing and is considered a more complex procedure. Both methods described above treat the temporal feature description of each pixel as a vector and do not delve into learning deep features from the temporal domain. Consequently, this limitation hampers further advancements in the accuracy of LULC classification.

Deep learning-based TSRSI classification has emerged as an active research area in recent years [28], [29]. Deep learning techniques excel at handling the complex characteristics of temporal signals and possess strong abilities for autonomous learning [30], [31], [32]. In comparison to traditional methods, deep learning approaches can automatically extract high-level features from raw data without manual intervention, thereby enhancing the accuracy of classification. Some deep learning models such as 1-D CNNs (1D CNNs), recurrent neural networks (RNNs), and long short-term memory (LSTM) networks have been successfully employed to extract features from time-sequential data, demonstrating significant improvements in classification performance [32], [33], [34]. However, the vectors received by these deep networks often exhibit certain correlations. If these correlations are not adequately considered during training, the model's training effectiveness may be compromised. To address this issue, some studies have incorporated self-attention mechanisms into deep networks to capture temporal or spatial relationships, thus achieving precise classification results [11], [28], [34], [35]. It is evident that capturing correlations between features is crucial for improving the classification of TSRSIs.

The above-mentioned studies aim to obtain the relationships between time-series features from the perspective of parameter training. Most of them focus on the temporal feature curves as the main research object [17]. However, due to the limitations of spatial representation, these methods can only obtain features from 1-D space, which may potentially lead to the overlooking of important features resulting from interactions between features. Recently, further studies have provided evidence that the rich information generated through certain mappings between features can enhance the attribute description of objects [35], [36]. Hence, we have reason to speculate that mapping the relationships between all features of individual pixels in a temporal image to 2-D or 3-D spaces, constructing matrices or tensors, can generate texture information and other spatial information that is more expressive than the representations provided by temporal feature curves. This allows us to fully leverage the learning advantages of CNNs in the spatial domain and obtain more abstract and stable features from 2-D matrices or 3-D tensors. Currently, CNN-based deep learning methods have achieved successful applications in scene classification of remote sensing images and hyperspectral image classification [37], [38]. However, mapping feature relations into 2-D matrices or 3-D tensors and using CNNs to learn deep features is an innovative topic.

In conclusion, learning deep features is a key factor in enhancing the classification performance of temporal remote-sensing images. From a feature learning perspective, establishing relationships between features and mapping temporal characteristics to 2-D and 3-D spaces allows for the extraction of more abstract deep features using CNN methods. Given this, the present study aims to address the challenges faced in temporal remote sensing image classification by investigating methods to extract deep features from feature relationships. Specifically, the proposed approach involves.

1) Developing a mapping mechanism based on the decomposition of temporal feature curves to describe feature relationships on matrices and tensors. This enables the transformation of feature relationships into 2-D and 3-D spatial texture information.
2) Designing a fusion approach that combines multiscale, 2-D, and 3-D CNNs using the spatial texture information of feature relationships. This facilitates the learning of deep features.
3) Constructing a classification framework for temporal remote sensing images based on feature relationship learning to further improve accuracy. The proposed method aims to uncover deeper features from temporal data and is expected to yield superior results compared to existing approaches. Therefore, it holds significant theoretical and practical implications for extracting high-precision and detailed LULC information from temporal remote sensing images.

II. Data Description

This study utilized the dataset introduced by Dou et al. [17] for experimental purposes. The dataset focuses on the Sutter County and Kings County regions in California, USA. Known for their extensive agricultural activities, these two regions offer a diverse range of crop types, rendering them ideal for conducting LULC classification using TSRSIs.

The survey data for this dataset was collected in 2014 through a collaborative effort between the California Department of Water Resources. They employed a combination of satellite imagery, high-altitude digital images, local data sources, and field surveys to generate accurate and reliable land use maps. This map was further processed to create 30 m grid data, which were then paired with Landsat images.

The dataset encompasses 20 distinct categories, primarily consisting of various crops, along with additional noncrop categories such as urban areas, wetlands, and other vegetation types (refer to Table I for the specific classification scheme) [17]. In terms of satellite imagery, the dataset comprises 19 Landsat 7 ETM+ images from the WRS path/row combinations 44/33 and 42/35, as well as 17 Landsat eight OLI images. To ensure data consistency, the ETM+ images utilized bands 1–5 and 7, while the OLI images employed

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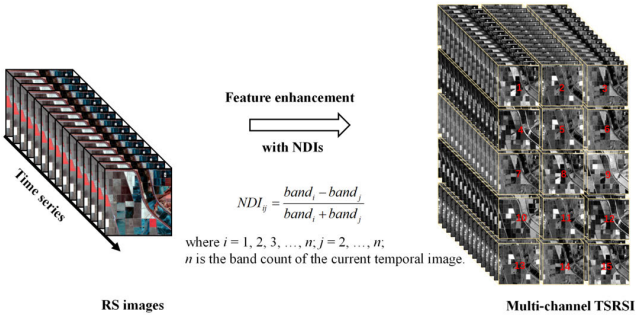


Fig. 1. Construction of NDIs-based multichannel TSRSI.

bands 2–7. This comprehensive collection of images results in a dataset with 36 different time points of surface reflectance data.

To enable fair comparisons of classification performance using novel methodologies, this study adheres to the original paper's approach of randomly partitioning the land cover samples into two sets. Specifically, 50% of the samples are allocated for training purposes, while the remaining 50% are dedicated to testing the performance of the proposed methods.

Research has consistently shown that leveraging various enhanced indices such as NDVI, enhanced vegetation index (EVI), and others for constructing time-series images offers distinct advantages over using raw data alone. The main rationale behind this is the ability of these indices to capture more pronounced temporal features. Building upon this understanding, Dou et al. [17] proposed a novel approach for constructing TSRSIs based on normalized difference indices (NDIs). This methodology involves applying NDIs to individual images within a given time period to enhance their features, resulting in a feature-enhanced image with 15 distinct channels. Subsequently, the corresponding NDI-enhanced images are stacked channel-wise to create 15 unique time-series images, as visually represented in Fig. 1.

Fig. 2 showcases the temporal feature curves of various land cover categories across different channels. It is evident that there is a strong discriminative capability between the different categories within each channel. In this study, we concentrate on the multichannel time-series images as the research subject, delving into the classification of TSRSI based on feature relationships.

III. Methods

A. Feature Relationship Construction

Mapping the relationships between features into texture patterns can enhance the distinguishability of land features from an alternative perspective. To achieve this, Dou and Zeng [36] projected the feature relationships onto a 2-D plane, generating a feature relationship graph that was effectively employed for hyperspectral remote sensing image classification. This approach yielded promising results. Since both time-series data and hyperspectral features are sequential in nature, we can leverage this characteristic to transform the temporal feature relationships into a 2-D plane. By doing so, we can explore methods for uncovering deep-level features

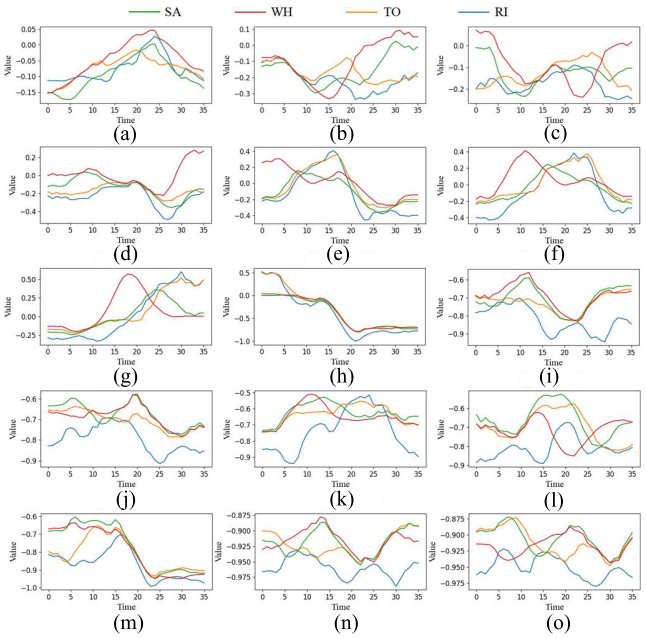


Fig. 2. Temporal feature curves of RI, TO, SA, and WH. (a) Chanel 1, (b) chanel 2, (c) chanel 3, (d) chanel 4, (e) chanel 5, (f) chanel 6, (g) chanel 7, (h) chanel 8, (i) chanel 9, (j) chanel 10, (k) chanel 11, (l) chanel 12, (m) chanel 13, (n) chanel 14, and (o) chanel 15.

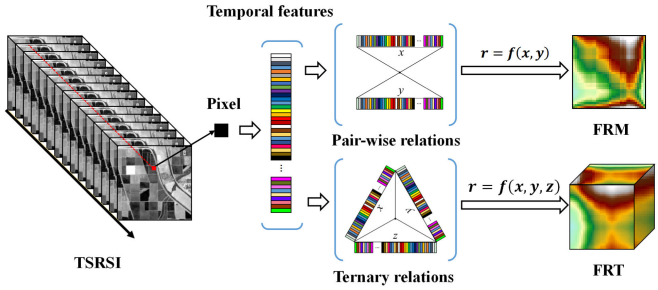


Fig. 3. Illustration of constructing individual pixel feature relationships in TSRS.

from the 2-D texture patterns. In addition, the relationships between features are not limited to pairwise associations; they can be extended to triplets, enabling the mapping of feature relationships into a 3-D data cube, represented as a 3-D tensor. This enables us to explore methods for learning deeper-level features from the 3-D texture patterns of feature relationships.

In light of this, the construction of feature relationships for each individual pixel in a TSRSI can be visualized as depicted in Fig. 3. To begin, the temporal features of the pixel are represented as a vector. Subsequently, pair-wise relationships are established by calculating the angles between the feature vectors, leading to the generation of a feature relationship matrix. Moreover, ternary relationships are formed, resulting in the creation of a feature relationship tensor (FRT). Considering that different land cover classes exhibit distinct temporal feature curves in the time-series image, the texture information encapsulated within the feature relationship matrix and tensor will naturally vary. This discrepancy can thus be employed as a vital basis for distinguishing between different land cover categories.

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TABLE I
CLASSES USED FOR CLASSIFICATION

Class label	Description	Training set		Testing set	
		Sutter	Kings	Sutter	Kings
RI	Rice	203081	0	217524	0
SU	Sunflowers	24530	0	18024	0
CSS	Corn, Sorghum and Sudan	9486	162568	11581	164171
AAM	Alfalfa and Alfalfa Mixtures	14122	64688	12994	65037
PN	Peaches/Nectarines	16422	134131	16812	131530
TO	Tomatoes	29669	80860	18563	103920
AL	Almonds	13979	10766	18208	10996
SA	Safflower	3633	72278	1459	68458
BD	Beans (Dry)	16030	47337	19100	54745
ID	Idle	95403	39124	101186	47086
MP	Mixed Pasture	7311	1430	6838	2450
PPA	Plums, Prunes and Apricots	40721	286506	35236	309840
WA	Walnuts	70948	3343	77280	3554
WH	Wheat	9698	7945	7800	6396
YP	Young Perennials	7936	40237	8814	38783
MW	Managed Wetland	26341	66741	7608	66576
UR	Urban	28353	1330	23265	1043
CO	Cotton	0	28152	0	20345
PI	Pistachios	0	60729	0	14713
OT	Other	19205	80272	18319	77118

B. Feature Relationship Mapping Method

The use of vectors to describe the features of pixels in TSRSI adds complexity to feature learning with CNNs. This is because the vector representation fails to capture the interrelationships between features, making it challenging to effectively express the rich information resulting from feature interactions. To address this limitation, in this article, we build upon the vector representation of temporal features and employ a mapping method to decompose the temporal feature curves and establish relationships between features. This approach enables the generation of matrices and tensors that can represent the relationships between every pair of features and every triplet of features, respectively. The specific method will be described in Sections III-B1 and III-B2.

1) Feature Relationship Matrix Construction: Fig. 4(a) depicts a simulated temporal feature curve of a pixel in a TSRSI. The curve includes a fixed point A and a moving point B, with an additional point C positioned in the middle of the temporal coordinate axis. As point B traverses the curve over time, it generates a curve illustrating the variation of angle ∠ABC with respect to the temporal sequence [as shown in Fig. 4(b)]. The fluctuation of angle ∠ABC reflects the relationship between the feature at point A and other features through the correlation method's mapping. This analysis also serves to decompose the original curve while exhibiting distinct temporal characteristics. Here, setting C as the midpoint can create a balance, preventing ∠ABC from becoming excessively obtuse or acute.

Based on this, by calculating the relationship between the feature at each point on the temporal feature curve and other features, different decomposition curves are obtained, and these curve values are stored in a matrix. The curves obtained from features at different time instances are distinct, while the curves from adjacent time instances exhibit gradual transitions. This allows the resulting feature relationship matrix to form specific textures [as shown in Fig. 4(c)].

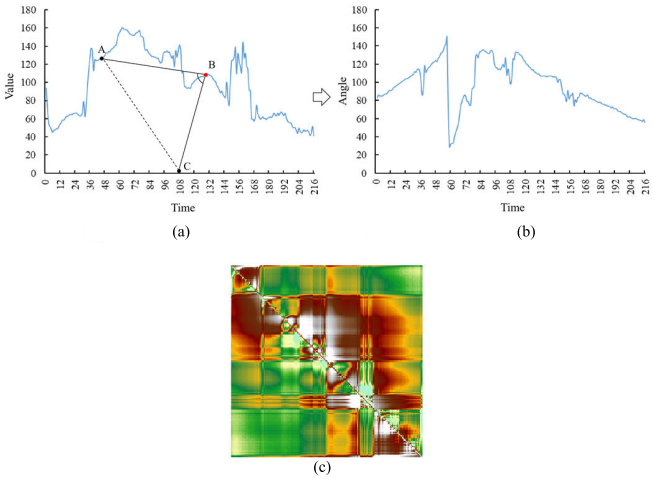


Fig. 4. Construction of the FRM based on angle calculation (the rows and columns in (c) represent the temporal positions of A and B, respectively. (a) Temporal feature curve, (b) temporal feature decomposition curve based on point A, and (c) FRM.

2) FRT Construction: As depicted in Fig. 5(a), there are three points, namely A, B, and C, on the temporal feature curve. The feature relationship between points A and B, as well as between points A and C, is established using the angle ∠BAC. By fixing point A and allowing points B and C to represent any two points on the temporal feature curve,

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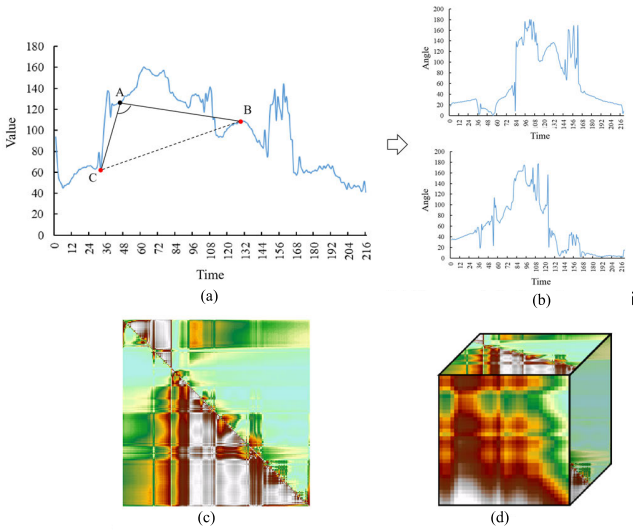


Fig. 5. Construction of the FRT based on angle calculation (the rows and columns in (c) represent the temporal positions of B and C, respectively; the rows, columns, and deeps in (d) represent the temporal positions of A, B, and C, respectively). (a) Temporal feature curve, (b) temporal feature decomposition curve based on point A, (c) FRM based on point A, and (d) FRT.

a matrix can be constructed containing all possible values of ∠BAC [as shown in Fig. 5(c)]. The variations in ∠BAC exhibit distinct temporal characteristics. In addition to reflecting the relationship between the feature at point A and any other two features, this matrix also converts the original temporal feature curve into corresponding texture information.

Based on this approach, the relationship between each feature on the temporal feature curve and any other two features can be calculated, resulting in different matrices. These matrices can be overlaid to obtain a tensor that contains corresponding texture information in different spatial directions [as illustrated in Fig. 5(d)].

C. Feature Learning Method

The CNN is a powerful deep-learning model specifically designed for processing images, videos, and other 2-D or 3-D data. It extracts features from input data through convolutional layers, reduces data volume, retains important information through pooling layers, and finally performs feature combination, classification, or recognition through fully connected (FC) layers [39], [40].

Traditional CNNs tend to prioritize global features and overlook local details as the number of convolutional layers increases, thus affecting the extraction of information at different scales [40], [41], [42]. To address these limitations, this article proposes a novel approach called the 2-D–3-D multiscale CNN (referred to as 2-D–3-D MSCNN), which combines both 2-D and 3-D information to learn more expressive features from the synthetic texture-like representation of FRMs and FRTs. The process is illustrated in Fig. 6: we construct a 2-D multiscale CNN (referred to as 2-D-MSCNN) and a 3-D multiscale CNN (referred to as 3-D-MSCNN) to extract features at different scales from the FRMs and FRTs, respectively, and then merge the output features to

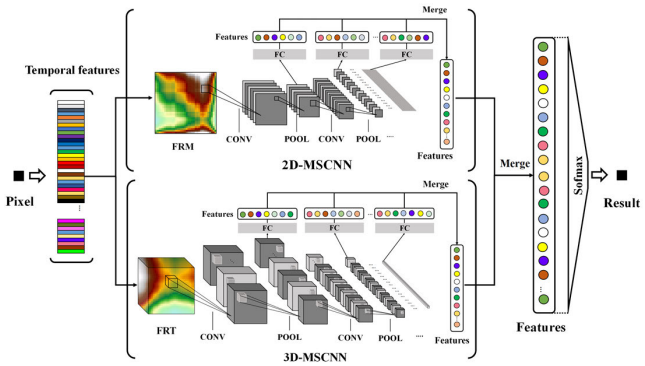


Fig. 6. Feature extraction based on 2-D–3-D MSCNN.

obtain a new set of fused features. Finally, the new features are fed into the Softmax layer for classification, resulting in the final class of the input pixel. This approach aims to overcome the limitations of traditional CNNs and enable more comprehensive feature learning.

The TSRSI constructed in this study is a multichannel image. Therefore, FRM and FRT are multichannel as well. The matrices and tensors included in them will be used for feature learning with the 2-D–3-D MSCNN. Afterward, the learned features will be combined to derive the ultimate outcomes of feature learning. Specifically, the methods for constructing the 2-D-MSCNN, 3-D-MSCNN, and feature fusion are described as follows.

1) Construction of 2-D-MSCNN: The construction process of 2-D-MSCNN, as depicted in Fig. 6, involves adding an FC layer after each pooling layer in the 2-D CNN (referred to as 2-D-CNN). These FC layers are then combined to form a new FC layer, which outputs the features learned by the 2-D-MSCNN.

2) Construction of 3-D-MSCNN: The 3-D-MSCNN extends the construction method of 2-D-MSCNN to three dimensions (as shown in Fig. 6). In addition to the 3-D CNN (referred to as 3-D-CNN), an FC layer is added after each pooling layer. Finally, all the FC layers are merged to create a new FC layer, which outputs the features learned by the 3-D-MSCNN.

3) Feature Fusion Method: Adding an FC layer after each pooling layer is primarily done to integrate and abstract the features extracted through convolution and pooling layers. This allows for the merging of features from different scales ultimately. To fuse the output features from the FC layers of 2-D-MSCNN and 3-D-MSCNN at different scales, a point-wise addition approach is employed. The calculation formula for the ith FC layer h^i is given by the following equation:

$$h^i = f(W^i x^i + b^i) \tag{1}$$

where $i = 1, 2, 3, \ldots, n$ (n represents the number of pooling layers), x_i denotes the unfolded features of the ith pooling layer, and W^i and b^i are the corresponding weight matrix and bias. The function $f(\cdot)$ represents the construction function for the FC layer.

The new FC layer h resulting from the fusion of all FC layers is denoted as (2) and its output represents the fused

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TABLE II
CONFIGURATION OF THE 2-D–3-D MSCNN

Configurations	Value
Number of units in FC layers	512
Number of filters in 2D-MSCNN	32
Number of filters in 3D-MSCNN	32
Filter size of 2D conventional layers	3×3
Filter size of 3D conventional layers	3×3×3
Pool size of 2D pooling layers	2×2
Pool size of 3D pooling layers	2×2×2
Number of layers in 2D-MSCNN	3
Number of layers in 3D-MSCNN	3

features obtained through different CNN scales

$$h = \sum_{i=1}^{n} h^i. \tag{2}$$

For the output feature vectors of the FC layers in 2-D-MSCNN and 3-D-MSCNN, a vector concatenation method is introduced. The two features are directly concatenated to serve as the input for the FC layer of the 2-D–3-D MSCNN.

Let h_{2d} represent the FC layer of the 2-D-MSCNN, h_{3d} represent the FC layer of the 3-D-MSCNN, and the output feature vectors of h_{2d} and h_{3d} are denoted as T_{2d} and T_{3d}, respectively. The fusion feature vector T is obtained by concatenating T_{2d} and T_{3d} as follows:

$$T = \text{concat}(T_{2d}, T_{3d}) \tag{3}$$

where concat(·) represents the feature concatenation function.

The FC layer is then computed as follows:

$$H = f(\text{TW} + b) \tag{4}$$

where W and b denote the respective weight matrix and bias, followed by the application of the ReLU activation function to obtain the fused feature vector P as follows:

$$P = \text{ReLU}(H). \tag{5}$$

Based on the description in Fig. 6, we designed three-layered convolutional networks for FRM and FRT, aiming to establish a 2-D–3-D MSCNN. The specific configurations are shown in Table II.

IV. EXPERIMENTAL RESULTS

A. Accuracy Evaluation

To validate the effectiveness of our proposed method, we conducted a comparison with the time series image classification framework (TSICF) proposed by Dou et al. [17]. Their framework comprises two main modules: the feature extractor and the classification model. The feature extractor is a deep learning architecture specifically designed to extract rich features from the temporal features of TSRSI. It can effectively learn and represent these features in a vector format. The resulting vector is further divided into several subsets, referred to as segments. The classification model consists of multiple classification networks, with each network corresponding to a segment. These networks are capable of producing class probabilities for the input data. To obtain the final classification results, the outputs from all the classification networks are integrated, enabling more accurate classification outcomes. Dou et al. [17] conducted experiments using various deep learning architectures, including one-dimensional simple CNN (1-D SCNN), temporal CNN (TCNN), Zhong's one-dimensional CNN (Z-1DCNN), and LSTM, to train their TSICFs. The results demonstrated good performance in classifying the time-series image data. Notably, the TCNN-based TSCF outperformed the other architectures, achieving the highest accuracy among them. Hence, we selected the TCNN-based TSICF as the comparative method for our study. The specific parameter settings can be found in [16].

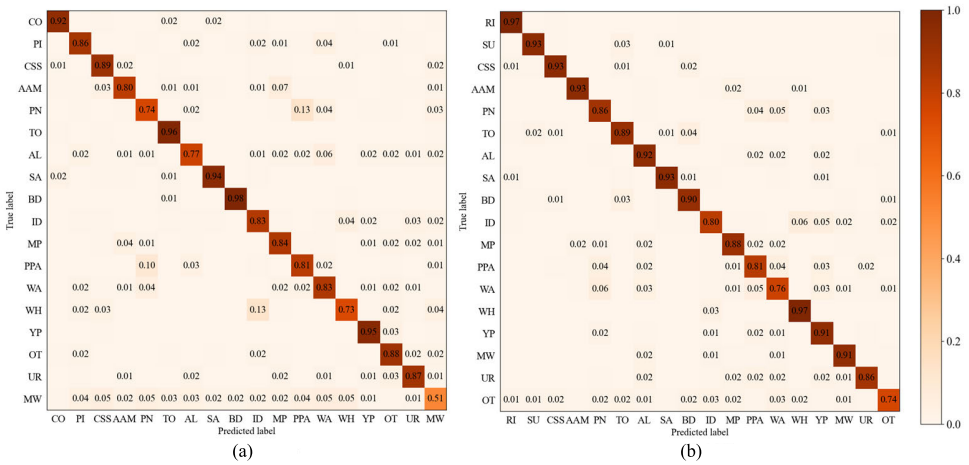
In addition, to evaluate the impact of the FRM and FRT on accuracy improvement, we performed classification using both 2-D MSCNN and 3-D MSCNN models. Furthermore, we conducted a comparative analysis by incorporating Dou's experimental results, which involved TCNN, Z-1DCNN, LSTM, GenELM, GB, and RF, all of which are used for temporal sequence classification, to compare against our classification results in this study [17].

The overall accuracy (OA) comparison of different classification methods is summarized in Table III. It is evident that our proposed novel approach has made significant advancements in improving accuracy. The 2-D–3-D MSCNN model demonstrates an accuracy improvement ranging from 3.26% to 5.12% compared to the TCNN-based TSICF. Furthermore, when compared to other benchmark methods, the accuracy is elevated by 9.82% to 16.3%. This signifies that the 2-D–3-D MSCNN model successfully learns deep-level features from the FRM and FRT, which contribute to improved accuracy. In contrast, the TCNN-based TSICF, despite employing an ensemble strategy, like other deep learning methods used for comparison, only learns from 1-D features and fails to effectively uncover the potential of newly generated features resulting from the interplay among different features. On the other hand, the 2-D MSCNN and 3-D MSCNN models learn deep-level features from pair-wise and ternary feature relationships, respectively, leading to higher accuracies compared to other benchmark methods. However, their accuracies, while relatively elevated, still fall short of the performance achieved by the 2-D–3-D MSCNN model (its error distributions are illustrated in Fig. 7). This suggests that simultaneously learning deep-level information from both pairwise and ternary feature relationships is essential to enhance the classification accuracy of TSRSIs.

B. Classification Results Mapping

The mapping accuracy of LULC is another crucial indicator that reflects the quality of classification algorithms. To this end, we utilized different algorithms to classify the Sutter and Kings datasets, and the results are presented in Figs. 8 and 9. It is evident that the employment of TCNN, Z-1DCNN, and LSTM led to serious misclassification phenomena, which

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(a) (b)

Fig. 7. Error distribution of the 2-D–3-D MSCNN. (a) Sutter. (b) Kings.

TABLE III
CLASSIFICATION ACCURACY ACHIEVED BY VARIOUS CLASSIFIER METHODS

Classification methods	Sutter		Kings	
	OA (%)	Kappa	OA (%)	Kappa
2D-3D MSCNN	88.63	0.866	83.26	0.811
2D MSCNN	86.39	0.841	78.15	0.752
3D MSCNN	87.26	0.853	81.33	0.789
TCNN-TSICF	83.51	0.805	79.97	0.772
TCNN	78.81	0.752	69.04	0.641
Z-1DCNN	76.22	0.721	70.03	0.650
LSTM	75.16	0.709	68.94	0.638
GenELM	75.64	0.720	71.21	0.664
GB	74.28	0.699	66.93	0.615
RF	75.46	0.712	69.44	0.645

resulted in lower accuracy of the final LULC mapping effect. TCNN-TSICF has a relatively higher mapping effect, with the misclassification rate apparently lower than the previous three methods, while in comparison to all the contrast methods, the 2-D MSCNN, 3-D MSCNN, and 2-D–3-D MSCNN based on FRM and FRT in this article have a better mapping effect. The misclassification of LULC types was significantly less than the comparing methods. Using deep learning based on feature relationships with FRM and FRT can obtain more representative features in the temporal information. Therefore, the 2-D MSCNN, 3-D MSCNN, and 2-D–3-D MSCNN based on FRM and FRT can accurately identify pixels while obtaining high-quality deep features, thus improving the mapping effect of LULC.

V. DISCUSSION

A. Separability of FRM and FRT

To obtain the temporal features corresponding to each sample in various datasets, experiments were conducted. Subsequently, for each sample class, the average value of all features was calculated to derive the corresponding temporal feature curve. Then, employing the angle decomposition method proposed in this article, the FRM and FRT were computed for each temporal feature curve.

The similarity among classes is a crucial indicator for assessing the quality of features. Higher similarity suggests that the utilized features have a weaker ability to differentiate between different LULC types, whereas lower similarity indicates stronger discriminative power. In this article, the cosine function is employed as a metric to measure similarity. A higher cosine value implies greater dissimilarity between classes, making them easier to distinguish. During the construction of FRM and FRT, the value range of temporal features may have an influence on the recognition ability across various classes, as the range affects the magnitude of angles and thereby influences the formation of spatial textures. To address this issue, we multiply the temporal features by a coefficient α to adjust the angle decomposition of the original features. Fig. 10 presents the impact of coefficient variations on the cosine mean. For FRM, the highest cosine mean of 0.161 is achieved when $\alpha = 200$. For FRT, the peak value of 0.326 is obtained when $\alpha = 3800$. Subsequently, as α continues to increase, the cosine mean exhibits a gradual decreasing trend. Finally, the maximum values obtained through testing are adopted as the coefficients for constructing the temporal features before building FRM and FRT.

This article includes the Sutter and Kings datasets, which consist of 15 different channels. Due to space constraints, we will focus on presenting the feature relationship matrices (FRMs) and tensors of different classes in the first channel using heat maps and contour plots. In Fig. 11, we observe the FRM of different classes. It is evident that the FRMs of different classes exhibit distinct synthetic texture-like representation variations, indicating their strong discriminability. Moving on to Fig. 12, we examine the FRT of different classes. From different perspectives, FRT provides a wealth of synthetic texture-like representation, allowing for clear visual differentiation between different categories. Compared to the 1-D temporal curves, FRM and FRT offer more intuitive 2-D and 3-D graphical information, making them better suited for distinguishing between different classes.

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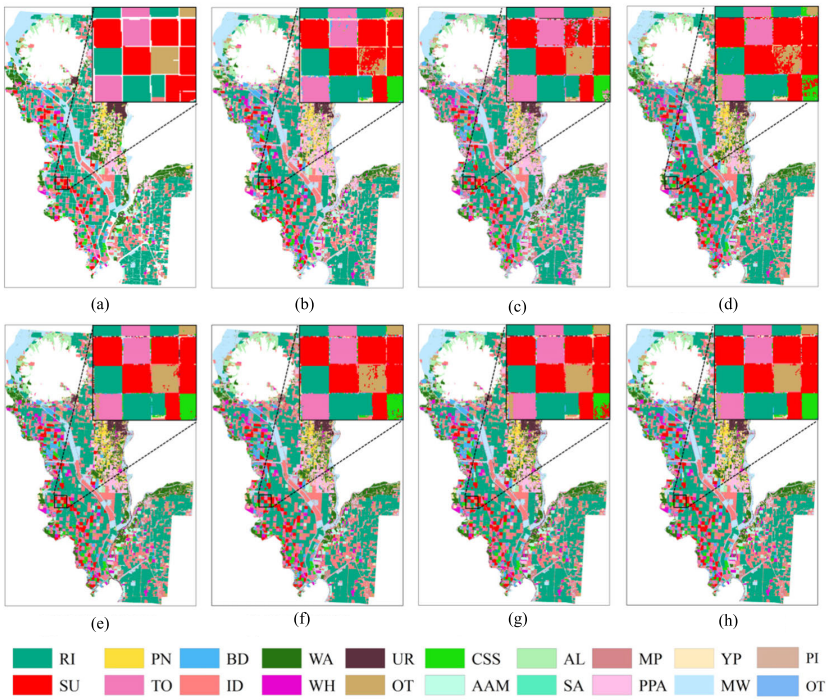


Fig. 8. Classification result mapping of the Sutter dataset with the use of different methods. (a) Ground truth, (b) TCNN, (c) Z-IDCNN, (d) LSTM, (e) TCNN-TSICF, (f) 2-D MSCNN, (g) 3-D MSCNN, and (h) 2-D–3-D MSCNN.

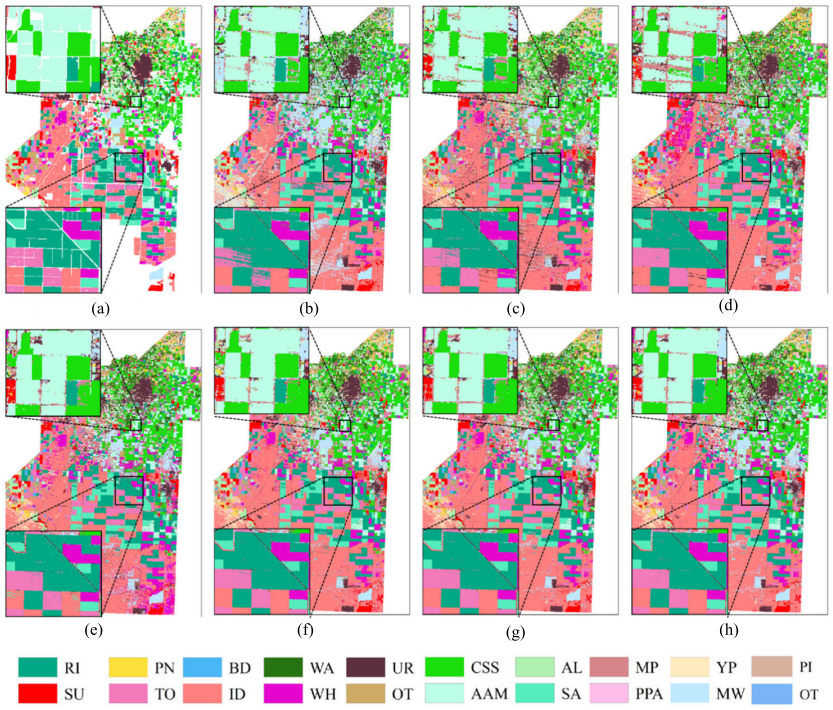


Fig. 9. Classification result mapping of Kings dataset with the use of different methods. (a) Ground truth, (b) TCNN, (c) Z-IDCNN, (d) LSTM, (e) TCNN-TSICF, (f) 2-D MSCNN, (g) 3-D MSCNN, and (h) 2-D–3-D MSCNN.

To quantitatively validate the effectiveness of FRM and FRT in distinguishing different classes, we employed the cosine function to analyze the similarity between different classes under various features. The similarity between classes is depicted in Fig. 13. It can be observed from Fig. 13(a) that when 1-D temporal features are utilized, there is high similarity between classes, indicating poor discriminability, with an average cosine similarity of only 0.013. However, with the implementation of FRM [Fig. 13(b)], the dissimilarity between classes is enhanced, and the average similarity increases to 0.136. When FRT is applied [Fig. 13(c)], the differences between classes become notably distinct, and the average similarity can reach 0.326. These findings further substantiate that compared to 1-D temporal feature curves, the extension

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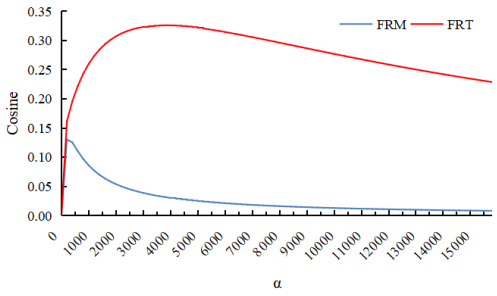


Fig. 10. Cosine similarity varies with the constant α.

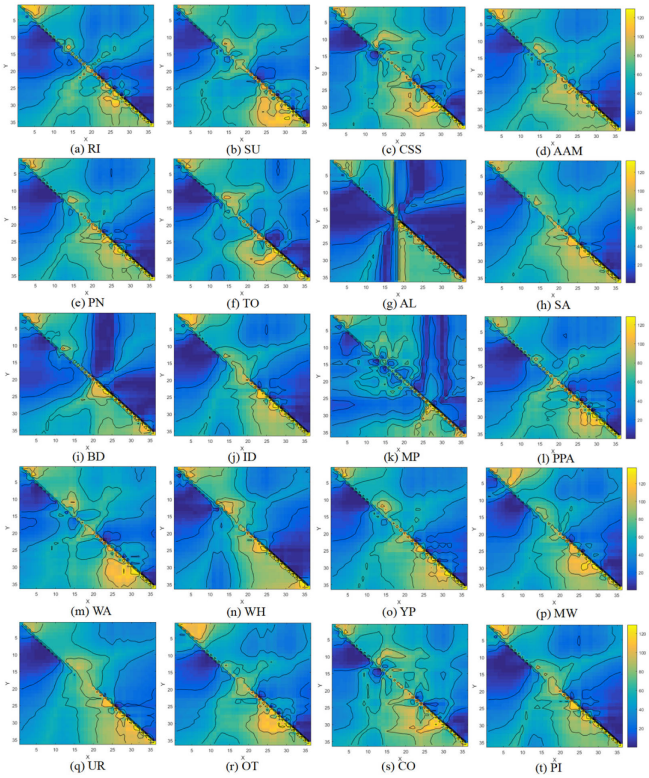


Fig. 11. FRM of different classes (x-axis represents column and y-axis represents row). (a) RI, (b) SU, (c) CSS, (d) AAM, (e) PN, (f) TO, (g) AL, (h) SA, (i) BD, (j) ID, (k) MP, (l) PPA, (m) WA, (n) WH, (o) YP, (p) MW, (q) UR, (r) OT, (s) CO, and (t) PI.

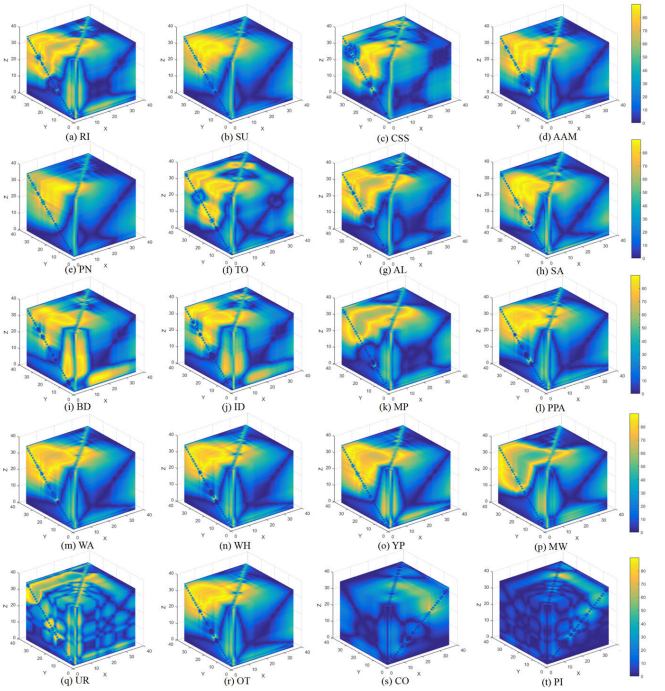


Fig. 12. FRT of different classes (x-axis represents column, y-axis represents row, and z-axis represents depth). (a) RI, (b) SU, (c) CSS, (d) AAM, (e) PN, (f) TO, (g) AL, (h) SA, (i) BD, (j) ID, (k) MP, (l) PPA, (m) WA, (n) WH, (o) YP, (p) MW, (q) UR, (r) OT, (s) CO, and (t) PI.

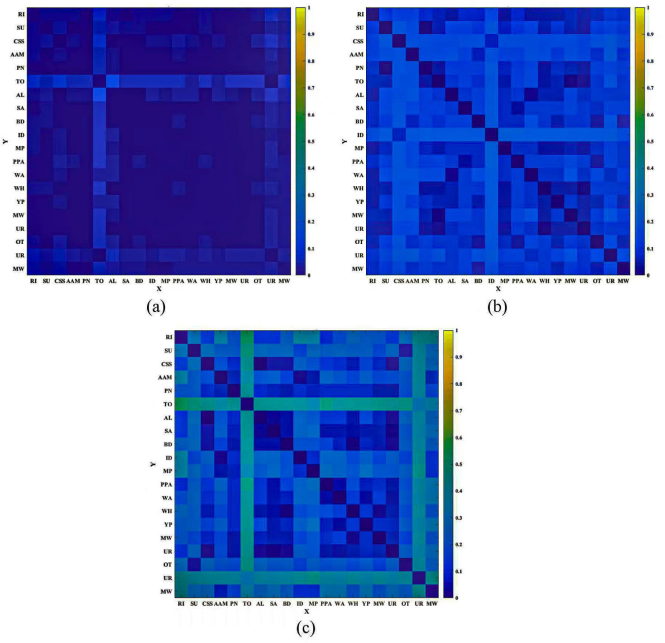


Fig. 13. Cosine similarity analysis of features for different classes uses (a) 1-D temporal features, (b) FRM, and (c) FRT (X, Y represent column and row).

to 2-D and 3-D space with FRM and FRT incorporates more information to enhance the differentiation between classes. This serves as a crucial aspect in boosting classification accuracy in this study.

In this study, a feature relationship mapping was constructed using an angle decomposition method. It should be noted that other methods can also be used to construct similar mappings. The key difference lies in the extent to which the generated spatial texture features are evident and can be used to distinguish between different categories. Here, we construct relationships by using the mean and difference between features, obtaining the corresponding FRM and FRT, respectively, and comparing their abilities to capture the similarity between classes. The results reveal that when the mean was used as the feature relationship, both FRM and FRT exhibited small differences between classes, with average cosine similarities of 0.008 and 0.009, respectively. On the other hand, when the difference was used as the feature relationship, the dissimilarity between categories increased for both FRM and FRT, with corresponding average cosine similarities of 0.022 and 0.024, respectively. It can be observed that utilizing the difference in feature relationships can better highlight the distinctions

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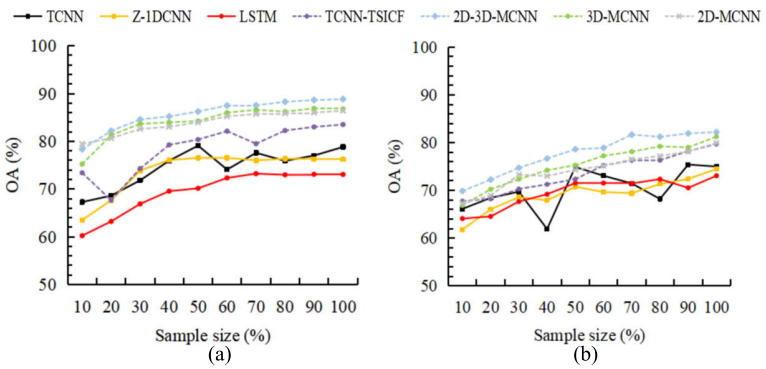


Fig. 14. Variations in accuracy as the sample size grows for the Sutter and Kings dataset. (a) Sutter. (b) Kings.

TABLE IV
CLASSIFICATION ACCURACY IMPROVEMENTS WITH USE OF DIFFERENT FEATURE RELATIONSHIP CONSTRUCTION METHODS

Relationships	Mean cosine similarity		Datasets	Classification accuracy with use of the proposed methods					
	FRM	FRT		2D MSCNN		3D MSCNN		2D-3D MSCNN	
				OA	Kappa	OA	Kappa	OA	Kappa
Mean	0.008	0.009	Sutter	79.88	0.759	76.06	0.714	80.45	0.766
			Kings	73.38	0.703	70.11	0.667	74.35	0.714
Difference	0.022	0.024	Sutter	82.68	0.793	83.01	0.797	84.55	0.815
			Kings	78.99	0.766	79.08	0.767	80.26	0.780
Proposed	0.136	0.326	Sutter	86.39	0.841	87.26	0.853	88.63	0.866
			Kings	78.15	0.754	81.33	0.775	83.26	0.811

between classes compared to employing the mean. However, these linearly obtained feature relationships still have lower discriminative capability for distinguishing between different classes as opposed to the feature relationship construction method proposed in this study. When applying the classification framework proposed in this study to the data generated using these two comparative methods, the results indicate that the feature relationship construction method proposed in this article has a greater advantage in improving classification accuracy (as shown in Table IV).

B. Classification Accuracy Improvements

1) Effects of Sample Size on Accuracy Improvement: The quantity of training samples plays a critical role in training classifiers [43], [44]. To investigate the effect of sample size on the proposed method, we incrementally increased the number of training samples by 10%. Fig. 14 demonstrates the variations in accuracy as the sample size grows for the Sutter and Kings datasets. It is evident that the accuracy of most methods increases with the addition of training samples. However, at an equivalent sample size, the 2-D–3-D-MSCNN consistently achieves higher accuracy compared to the comparative methods. Observing the changes in the accuracy curve, it is apparent that as the sample size continues to increase, the accuracy of the new method maintains an upward trend. The comparative methods used in this study usually rely on temporal feature curves as the raw input, making it challenging to distinguish between similar features since the relationships between features cannot be utilized during feature learning. Therefore, to enhance the model's ability to differentiate such features, the sample quantity must be increased. Constructing multinetwork ensemble models such as TCNN-TSICF is also a way to improve classification accuracy, but these operations inevitably lead to increased execution efficiency and complexity of the model. In contrast, the proposed FRM and FRT offer more expressive capabilities than the original temporal curves. Consequently, when utilizing 2-D–3-D-MSCNN, more abstract and representative features can be obtained, leading to improved classification accuracy at any training sample size.

2) Classification Improvements at Per-Class Level: The above analysis indicates that the 2-D–3-D-MSCNN, based on feature relationships, achieves higher accuracy compared to the contrastive methods. Fig. 15 demonstrates the accuracy of each category obtained using different approaches. It is evident that, in most cases, the 2-D–3-D-MSCNN exhibits a significant advantage in improving accuracy at per-class level. This suggests that the OA improvement of the 2-D–3-D-MSCNN is built upon the foundational enhancement of individual class accuracy; the proposed method does not suffer from overfitting, where high-accuracy classification results are disproportionately focused on specific classes. However, there are a few exceptions, such as the WA and ID in the Sutter dataset, where the 2-D–3-D-MSCNN's class accuracy is lower than that of the compared methods. These occasional outliers are reasonable and prevalent since multiclass classification inherently involves inevitable instances of confusion between different categories, thereby limiting

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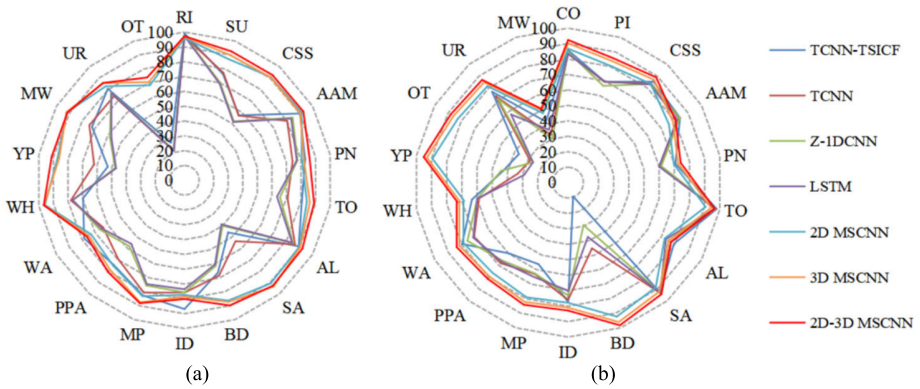


Fig. 15. Classification improvements at per-class level. (a) Sutter. (b) Kings.

TABLE V
TIME CONSUMPTION OF DIFFERENT METHODS (HOUR)

Method	Sutter dataset		Kings dataset	
	Training	Classification	Training	Classification
TCNN-TSICF	2.671	0.742	2.95	1.029
TCNN	0.143	0.284	0.206	0.594
Z-1DCNN	0.147	0.263	0.265	0.558
LSTM	0.162	0.250	0.277	0.502
2D MSCNN	0.187	0.356	0.301	0.611
3D MSCNN	2.813	0.783	3.156	1.131
2D-3D MSCNN	3.212	0.899	3.432	1.303
2D-3D-MSCNN (optimized)	2.681	0.758	2.816	1.098

the complete advancement of accuracy in certain classes. Simultaneously, this also highlights that no single classifier can be deemed flawless; different classification methods display varying advantages for distinct classes. Thus, within the research process of remote sensing image classification, it is crucial to concentrate on enhancing the accuracy of each class as much as possible, thereby achieving an overall improvement in accuracy.

3) Time Costs: The experimental setup utilized for this study involved a computer configuration with "64 GB RAM memory, i9-10900K 3.7 GHz processor, and NVIDIA TITAN RTX 24GB GPU graphics card." All programs were written in Python. Table V provides the training and classification times for the methods discussed in this article. Overall, training deep learning models is a time-consuming process. The introduced FRM and FRT methods in this study add additional time costs during the construction phase, resulting in longer training and classification times for the 2-D–3-D-MSCNN compared to the contrasting methods.

To minimize time consumption and enhance model runtime efficiency, we parallelized the construction of FRM and FRT using GPU. Table V also presents the time consumption of the 2-D–3-D-MSCNN before and after optimization, demonstrating an improvement of around 16% in runtime efficiency after optimization.

While the construction of FRM and FRT may introduce additional time costs, the optimization and parallelization using GPU have allowed us to enhance the runtime efficiency of the 2-D–3-D-MSCNN method, leading to improved performance and accuracy. This development holds significance for the utilization and advancement of deep learning models.

Although the new methods may take more time to execute, they provide enhanced performance and accuracy. In the field of deep learning, there is often a trade-off between time and precision. Despite the increased time costs resulting from parallelization and optimization to improve the models runtime efficiency, the sacrifice allows us to obtain superior results and higher accuracy, which are vital for numerous applications.

VI. CONCLUSION

The relationship between temporal features is mapped into spatial textures, allowing for the learning of deeper features and improving classification accuracy. This article introduces the construction of FRM and FRT using angular relationships between temporal features to generate 2-D and 3-D texture information, significantly enhancing the separability of different classification objects. High-quality useful features are learned using 2-D MSCNN and 3-D MSCNN. Subsequently, by combining FRM and FRT, feature learning and classification are conducted using 2-D–3-D MSCNN, resulting in a substantial improvement in the classification accuracy of TSRSI. Due to its more robust feature learning ability, the new method can achieve higher accuracy than comparative methods in most cases. Unlike existing learning methods, the new method learns from feature relationships rather than directly from spectral or spatial characteristics, enabling it to fully utilize all original features without the need for any data dimensionality reduction, thereby obtaining more intuitive

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features, improving classification accuracy, and enhancing the mapping quality of LULC.

The new method demonstrates excellent performance in the classification of TSRSIs. However, there are still aspects that need improvement. For example, in the experiments, we used angles to establish the relationships between temporal features, and the construction of feature relationships should be more diverse. To achieve higher accuracy, it is necessary to utilize diverse feature relationships to construct FRM and FRT in future research. In addition, FRM, especially FRT, has the drawback of being time-consuming in the construction process, therefore, the program code still needs further optimization to accelerate its running speed in future research.

In summary, the new method proposed in this article enhances the classification accuracy of TSRSIs by utilizing feature relationships. This framework is flexible and convenient and is expected to be applicable to more complex RS image classifications in future research.

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